LESSON 9-2 · CIRCLES

Choose Your Route: Practice

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

Three routes, one destination. Start anywhere on pages 1 and 2, move between routes whenever you like, and finish with page 3. A route names the task, not you.

PRACTICE · SECURE THE METHOD

Cards 1 to 3: write the center and the radius. Cards 4 and 5: write the equation. Card 6: complete the square, then find the center and radius. Then sketch card 1 on the grid.

1
Center and radius
$(x-2)^2+(y+3)^2=16$

2
Center and radius
$(x+4)^2+(y-1)^2=49$

3
Center and radius
$x^2+y^2=36$

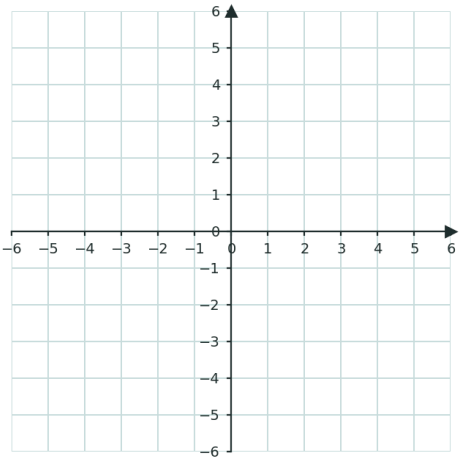
4
Write the equation
center $(-1,5)$, $r=3$

5
Write the equation
center $(0,-2)$, diameter 10

6
Complete the square
$x^2+y^2-6x-8y+9=0$

Working space. Flip the signs inside the brackets for the center. The number on the right is r^2, so take its square root.

Grid. Sketch card 1: the center, the radius and four points on the circle.



FIND THE ERROR

A student completed the square and read off the center and radius. Say what went wrong and write the correct center and radius.

$x^2+y^2+4x-6y-12=0 \Rightarrow (x+2)^2+(y-3)^2=12$

so the center is $(2,-3)$ and $r=\sqrt{12}$

Done when... the center is read with the signs flipped, the right-hand side is treated as r^2, and the numbers added in completing the square go on both sides.

LESSON 9-2 · CIRCLES

Choose Your Route: Apply and Investigate

Algebra II · Choose Your Route

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APPLY · USE IT IN CONTEXT

1. A model circular plaza in the Hira Cultural District has center the origin and diameter 60 m. A straight path along $y = 18$ crosses it. Where does the path enter and leave, and how long is it?

2. A model radio tower at $(2, -1)$ covers a radius of 5 km. Is a village at $(5, 3)$ inside, outside or on the edge of the covered area?

3. A model lake is the circle $x^2 + y^2 - 10x + 4y + 13 = 0$ (km). Find its center and radius.

4. A model path $y = x + 2$ meets the boundary $x^2 + y^2 = 34$. Find the two points where it does.

Done when: the center is placed first, the radius comes from r^2, and each answer says what it means in the context.

INVESTIGATE · FIND OUT WHY

1. Use the distance formula to derive the equation of the circle with center (h, k) and radius r.

2. Show that the line $y = 3x + 10$ is tangent to $x^2 + y^2 = 10$ by solving the system. Find the point of tangency.

3. For $x^2 + y^2 = r^2$ and $y = mx + c$, show that the line is tangent when $c^2 = r^2(1 + m^2)$. Check it with part 2.

Done when: your argument starts from the distance formula or from a single repeated solution, and every step stays balanced.

SPOTLIGHT · AN IDEA THAT IS MY OWN

A route nobody else used, a question nobody else asked, or a surprise you found:

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Which Route? Judge the Claims

Algebra II · Choose Your Route

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COMPARE THE ROUTES

Find the center and radius of $x^2 + y^2 - 6x + 4y - 3 = 0$ three ways. Then compare: which was quickest, and which would you trust?

ROUTE	my working	steps	where a slip could happen
Complete the square			
Center formula $\left(-\frac{D}{2}, -\frac{E}{2}\right)$			
Graph or technology			

My choice and the evidence for it:

JUDGE THE CLAIMS

Decide true, false or sometimes true. One counterexample is enough to show a claim is false; a few examples are not enough to show it is true.

A	**A line and a circle always meet at two points or not at all.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I test a line that just touches the circle?
B	**If the distance from the center to a line is less than the radius, the line meets the circle at two points.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I measure the perpendicular distance, not the distance to a point?
C	**The equation $x^2 + y^2 = -4$ is a circle of radius 2.**	☐ true ☐ false ☐ sometimes	*My evidence:* *Assumption I checked:* Did I check that the right-hand side is positive?

WHAT IF...? PREDICT FIRST, THEN TEST

1. What if the right-hand side is 0, as in $(x-1)^2 + (y+2)^2 = 0$? Predict the graph, then test with a point.

2. What if the radius doubles and the center stays? Predict what happens to r^2 and to the area. Then test.

3. Write a What if...? question of your own about a circle, predict the answer, and test it.

Done when... you have compared three routes with evidence, judged each claim with an example or counterexample, and tested at least one What if...? idea.